Linux Server Administration: command cheat sheet

AlmaLinux / RHEL family and Ubuntu / Debian. Use sudo where you are not root. Test configuration before every reload.

Find your way

`cat /etc/os-release`	Which distro and version
`pwd / ls -la / cd -`	Where am I, what is here, go back
`find /etc -name '*.conf'`	Find files by name
`grep -rn 'Listen' /etc/httpd/`	Search inside files
`less file / tail -f log`	Read a file / follow a log
`man cmd / cmd --help`	Help

Files and permissions

`cp -a src dst / mv / rm -r`	Copy keeping permissions, move, delete
`chmod 640 file / chmod g+w`	Numeric / symbolic permissions
`chown user:group file`	Change owner and group
`chmod 2770 /srv/project`	Shared folder: setgid keeps the group
`ls -ld dir / ls -Z`	Directory itself / SELinux labels
`umask`	Default permissions for new files

Users and sudo

`useradd -m -s /bin/bash tobi`	New user with home folder
`usermod -aG wheel tobi`	Add to group (-a: keep others!)
`id tobi / groups`	Who is in which group
`sudo -i / visudo`	Root shell / edit sudoers safely

Packages

`dnf upgrade | apt update && apt upgrade`	Install updates
`dnf install X | apt install X`	Install a package
`dnf provides /usr/bin/dig`	Which package has this file
`dnf history / dnf history undo N`	Review / undo a transaction
`dnf needs-restarting -r`	Reboot needed? (AlmaLinux)

Services and logs

`systemctl status|restart|reload X`	Control a service
`systemctl enable --now X`	Start now and at boot
`systemctl --failed`	Anything broken?
`journalctl -u X -n 50 -f`	Last 50 lines of X, then follow
`journalctl -p err -b`	Errors since boot
`systemctl daemon-reload`	After editing unit files

Processes and resources

`ps aux | grep [n]ginx`	Find a process
`top / htop`	Live view (M: memory, P: CPU)
`kill PID / kill -9 PID`	Ask to stop / force (last resort)
`uptime / nproc / free -h`	Load vs cores / memory
`df -h / df -i / du -sh *`	Disk space / inodes / folder sizes

SSH

`ssh-keygen -t ed25519`	New key (set a passphrase)
`ssh-copy-id user@host`	Install your public key
`sshd -t && systemctl reload sshd`	Test, then reload
`sshd -T | grep -i password`	Settings actually in force

Network

`ip a / ip r`	Addresses / routes
`ss -tulpn`	What listens on which port
`curl -I https://site`	HTTP headers only
`dig +short name A`	DNS lookup
`nc -zv host 22`	Is a TCP port open?

Firewall

`firewall-cmd --list-all`	AlmaLinux: what is allowed
`firewall-cmd --permanent --add-service=https`	Allow a service...
`firewall-cmd --reload`	...then apply
`ufw allow OpenSSH; ufw enable`	Ubuntu: SSH FIRST, then enable
`ufw status verbose`	Ubuntu: what is allowed

Web and HTTPS

`nginx -t && systemctl reload nginx`	Test, then reload Nginx
`apachectl configtest`	Test Apache config
`certbot --nginx -d site`	Get a certificate
`certbot renew --dry-run`	Prove renewal works

Database

`sudo mysql`	Admin login via socket
`mysqldump --single-transaction db | gzip > db.sql.gz`	Consistent backup
`gunzip < db.sql.gz | mysql newdb`	Restore into a test DB

SELinux (AlmaLinux)

`getenforce`	Enforcing? Keep it on
`restorecon -Rv /var/www`	Fix labels after mv
`setsebool -P httpd_can_network_connect_db on`	Allow web app -> remote DB
`ausearch -m AVC -ts recent | audit2why`	Why was it blocked?